



ULAB
UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Project Report

Course Title: Database
Management System Lab

Course Code: CSE 2302
Section: 1

Submitted to:

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Title:

University Resource Management System (URMS)

Introduction:

The University Resource Management System (URMS) is a dynamic web-based platform designed to streamline, automate, and digitize various administrative and academic processes within a university environment. The system offers a centralized interface for students, faculty, and administrators, reducing manual effort and enhancing operational efficiency. It eliminates dependency on paperwork and provides instant access to academic data. By incorporating real-time updates, role-based access control, and an intuitive interface, URMS aims to replace outdated legacy systems and serve as a scalable solution adaptable to institutional growth and evolving user needs.

Problem Statement:

Many universities continue to rely on manual data entry or fragmented software systems for student management, which results in inefficiencies, data redundancy, and communication breakdowns. Critical academic processes like grade publishing, billing, and course registration often suffer delays. Moreover, faculty members experience challenges managing course results and engaging with students due to a lack of streamlined tools. Without proper user authentication and centralized recordkeeping, these institutions are also at risk of data inconsistency and security breaches. URMS addresses these critical gaps through an integrated digital system that improves data flow, transparency, and operational accuracy.

Motivation:

The motivation behind developing URMS is driven by the vision of creating a smarter academic infrastructure that responds to modern educational challenges. With increasing student populations and complex course structures, managing university operations manually is neither scalable nor reliable. Our firsthand experience with delayed results, lack of student self-service portals, and administrative bottlenecks inspired the need for a comprehensive solution. URMS offers an opportunity to bridge the technological gap in institutional management while promoting digital literacy, transparency, and accessibility for all stakeholders involved in the academic ecosystem.

Objectives:

1. To develop a centralized web-based system for managing university operations.
2. To provide role-based access control for students, faculty, and administrators.
3. To allow students to view grades, billing information, and perform course preregistration.
4. To enable faculty members to submit course results and manage assigned courses.
5. To equip administrators with tools to manage users, assign courses, and monitor activity logs.
6. To implement real-time data updates using AJAX for smoother user interactions.
7. To ensure data integrity and security by using session management and input validation.
8. To build a responsive user interface that works across desktops, tablets, and smartphones.
9. To structure the system modularly for easy maintenance and scalability.
10. To enhance transparency, accountability, and reduce manual workload within academic workflows.

Background Theory:

URMS is built upon foundational principles of web development and database management systems (DBMS). The system utilizes the three-tier architecture model, comprising the Presentation Layer (frontend UI), Business Logic Layer (server-side logic in PHP), and Data Layer (MySQL). It incorporates the Model-View-Controller (MVC) concept for code organization, session management for security, and asynchronous for non-blocking data updates. These theories support the creation of a responsive, real-time user experience. Furthermore, it draws from the concept of CRUD operations, relational schema design, user-centered interface design, and role-based access control (RBAC) for function-specific permissions.

Target User and Use Case:

Target Users:

1. Students: View results, track billing information, and perform online course registration.
2. Faculty: Manage courses, upload student results, and view assigned course lists.
3. Administrators: Add/edit/delete users, manage course lists, and monitor system data.

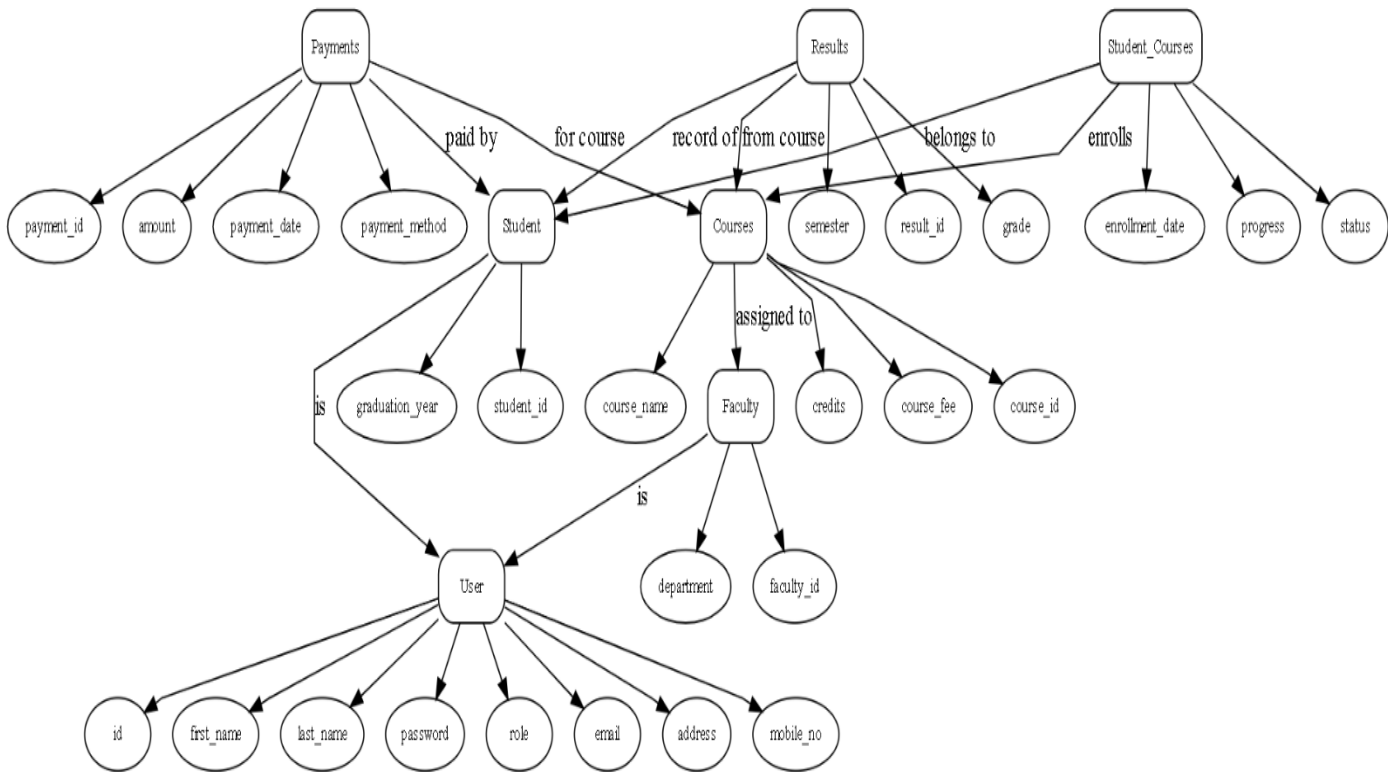
Use Case Scenarios:

1. A student logs in to check billing dues and course results for the semester.
2. A faculty member uploads grades for their assigned courses using a secure interface.
3. An administrator adds a new course, assigns a faculty member to it, and updates student enrollments.

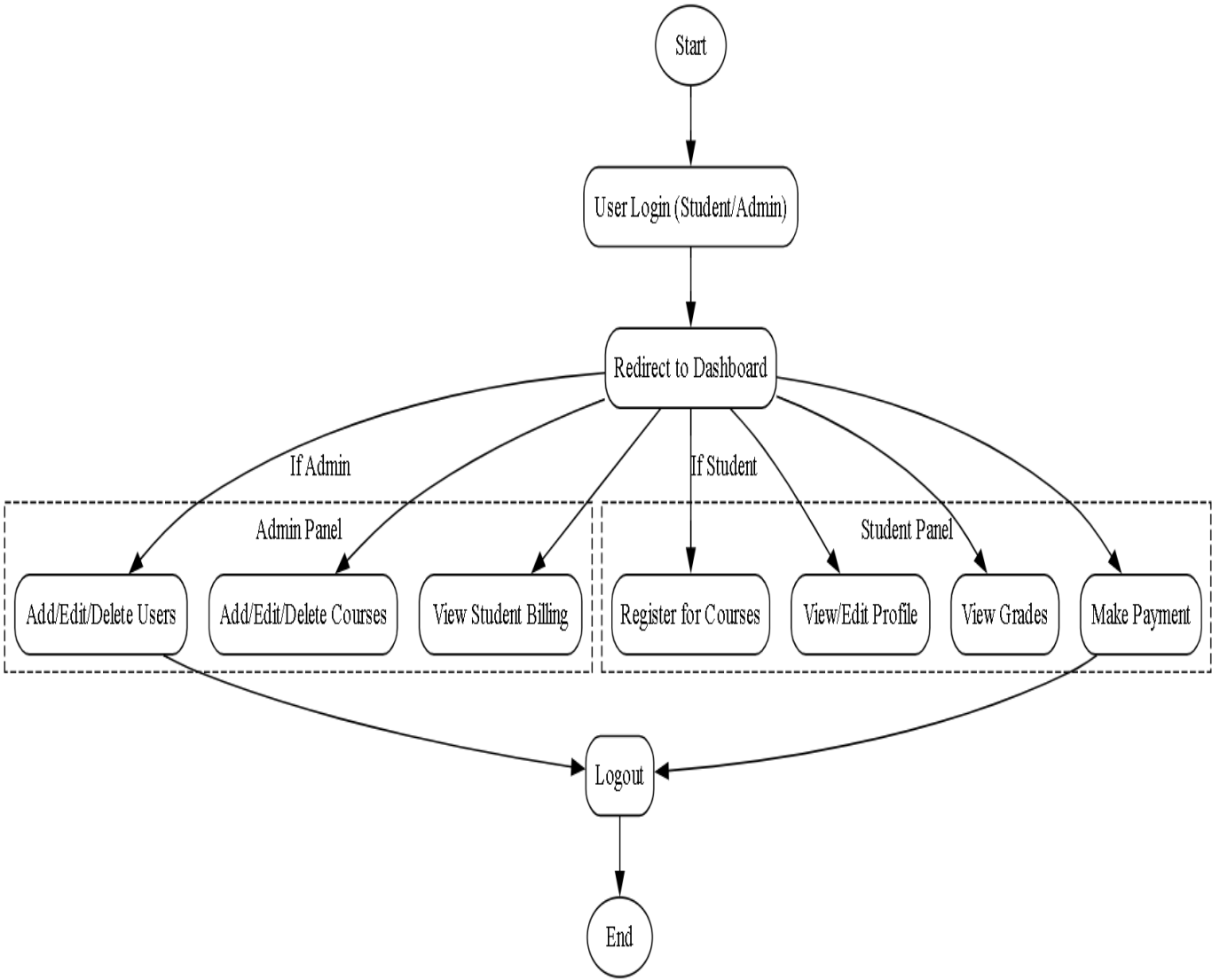
Features and Functionalities

- Secure login and session management using PHP.
- Role-based dashboard access with specific features for each user type.
- Real-time updates and data sync using AJAX.
- Course registration system for students with automated capacity checks.
- Faculty result upload module with validation and edit capability.
- Admin dashboard for user and course management with analytics insight.
- Billing system showing outstanding balances, due dates, and payment status.
- Logout mechanism and session timeout for improved security.
- Modular code structure for easy maintenance and future enhancements.

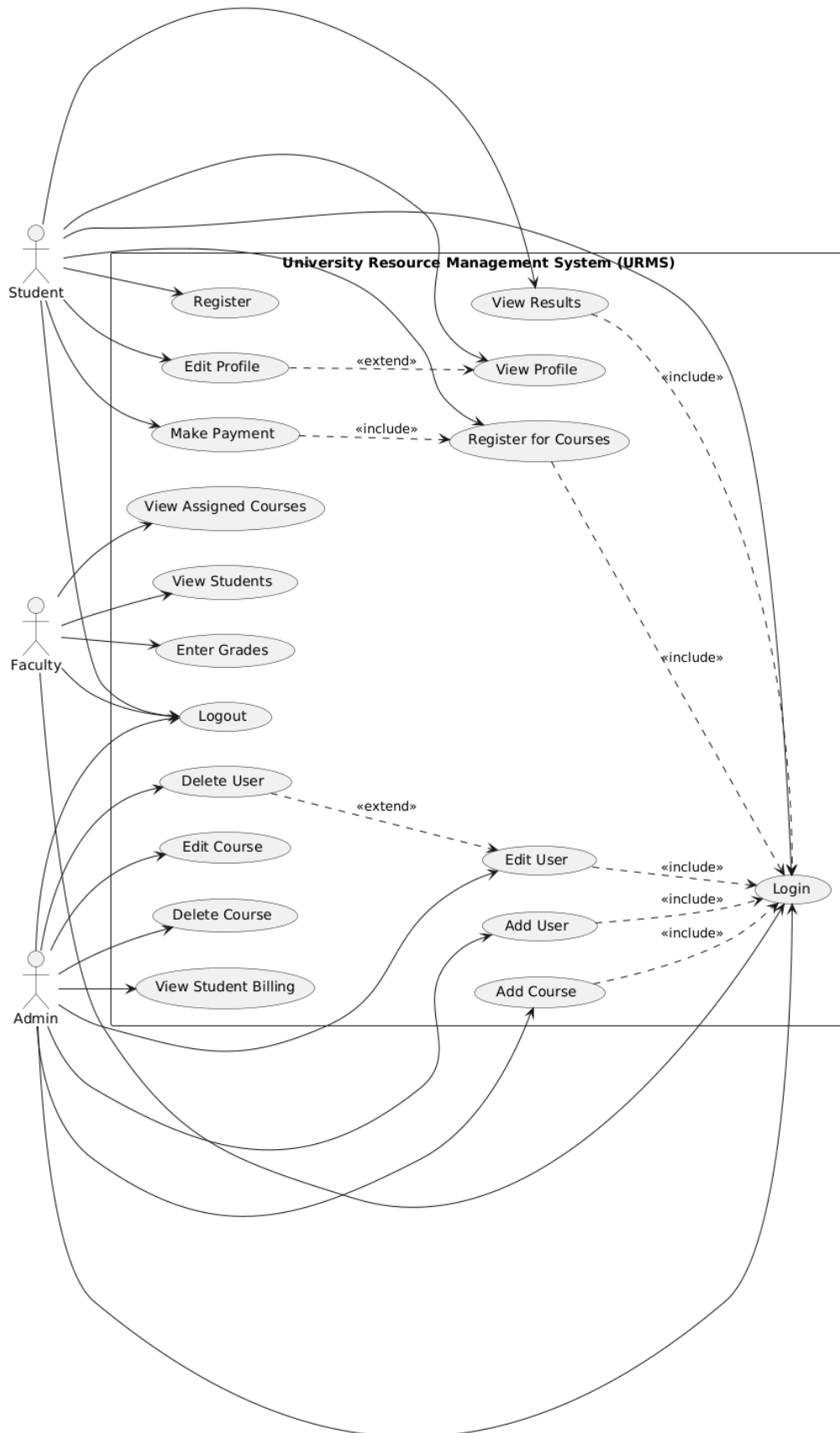
Entity-Relationship Diagram (ERD)



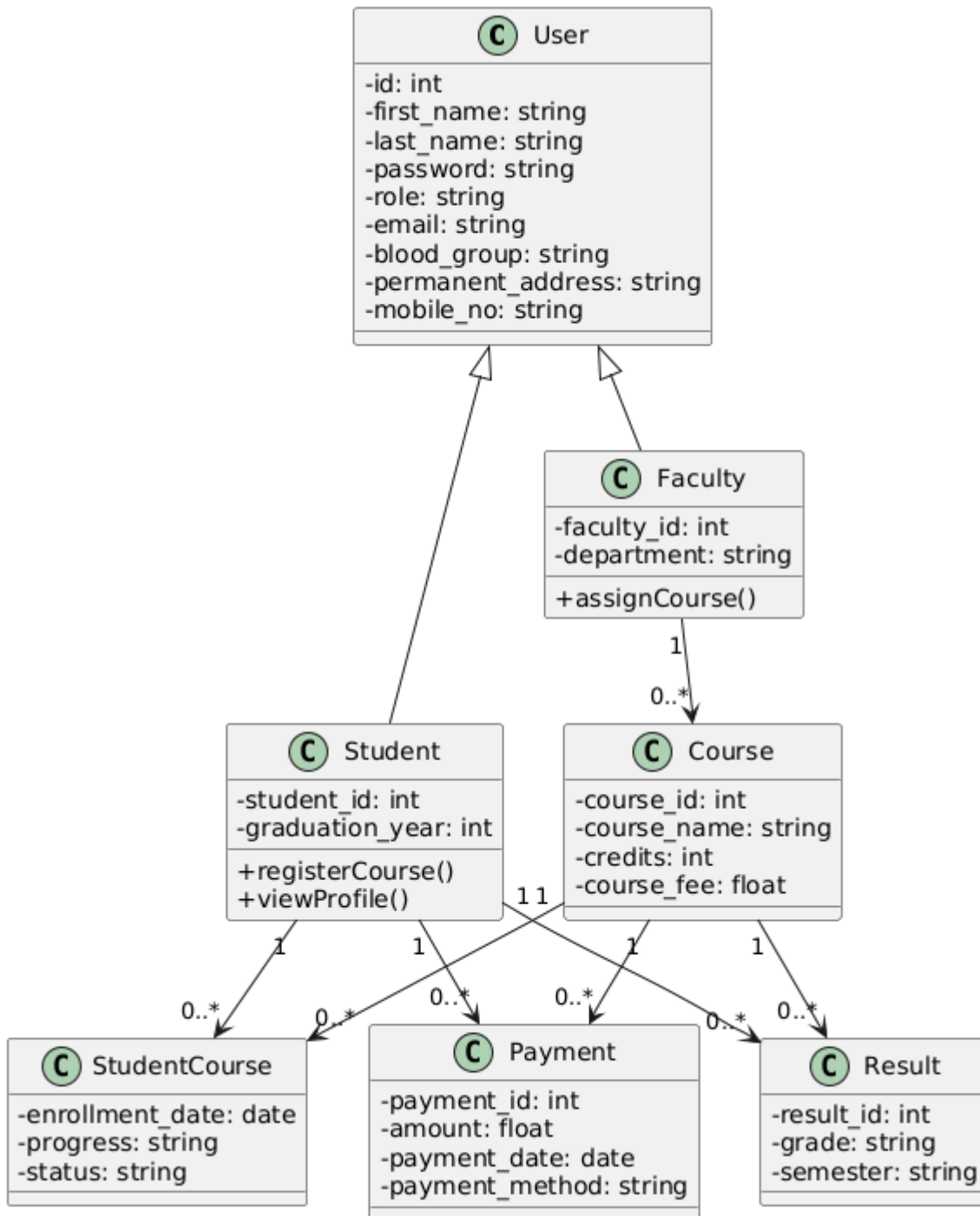
Flowchart:



Use Case diagram

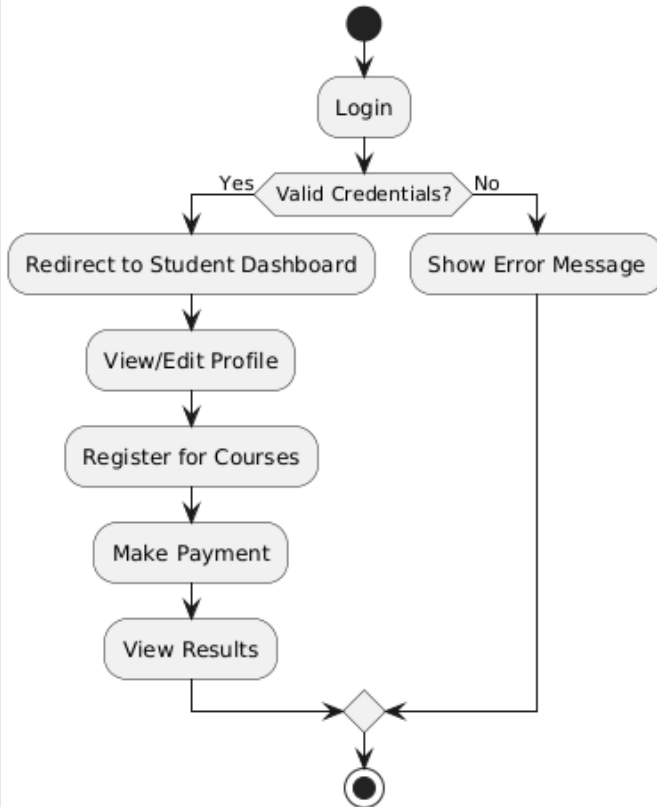


Class diagram

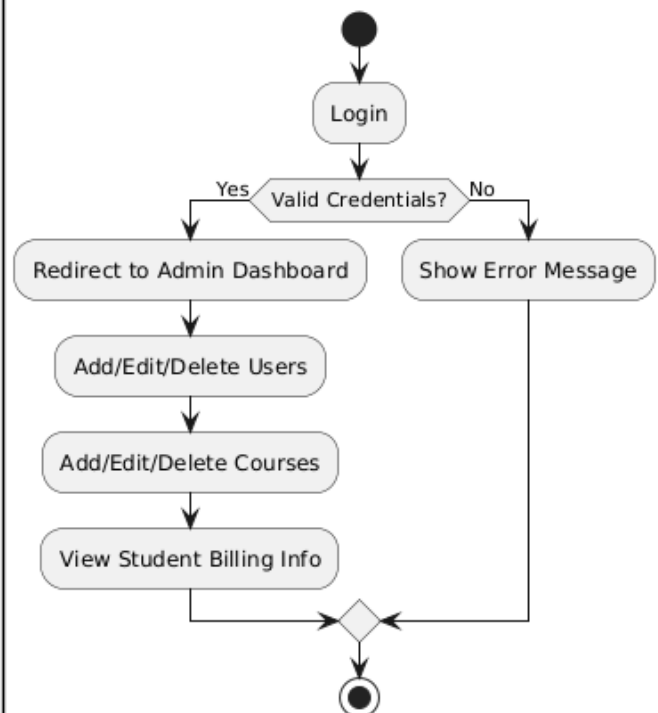


Activity diagram

Student



Admin



Technology:

1. **Frontend:** HTML for structure, CSS for styling, Bootstrap for responsive design
2. **Backend:** PHP for server-side scripting and data handling
3. **Database:** MySQL for storing structured data using relational tables
4. **Web Server:** Apache, run locally via XAMPP for development and testing

Methodology:

This project followed the Waterfall Software Development Life Cycle (SDLC) model. The stages included:

1. Requirement Gathering – Interviews with students and faculty for user needs
2. Design – Creating flowcharts, wireframes, and database schemas
3. Implementation – Coding the modules in PHP, HTML, JS, and CSS
4. Testing – Unit testing, role testing, and bug fixing
5. Deployment – Hosting locally in XAMPP, with configuration for a production-ready environment
6. Maintenance – Updating modules based on test feedback

SQL Queries Implemented

Basic SELECT Query:

```
SELECT * FROM users WHERE first_name = ?;
```

Get User by ID:

```
SELECT * FROM users WHERE id = ?;
```

Insert User:

```
INSERT INTO users (first_name, last_name, password, role, email, blood_group,  
    permanent_address, mobile_no)  
VALUES (?, ?, ?, ?, ?, ?, ?, ?);
```

Delete Course:

```
DELETE FROM courses WHERE course_id = ?;
```

Delete User:

```
DELETE FROM users WHERE id = ?;
```

Update Course Info:

```
UPDATE courses SET course_name = ?, credits = ?, course_fee = ?, faculty_id = ? WHERE  
course_id = ?;
```

View Creation:

```
CREATE VIEW student_course_view AS  
SELECT users.first_name, courses.course_name  
FROM users  
JOIN student_courses ON users.id = student_courses.id  
JOIN courses ON student_courses.course_id = courses.course_id;
```

Insert into Select Query:

```
INSERT INTO billing (student_id, amount)  
SELECT student_id, 5000 FROM student_courses;
```

Trigger:

```
CREATE TRIGGER after_user_insert  
AFTER INSERT ON users  
FOR EACH ROW  
INSERT INTO logs(activity)  
VALUES (CONCAT('New user added: ', NEW.first_name));
```

Stored Procedure:

```
DELIMITER $$  
CREATE PROCEDURE GetStudentBilling(IN stud_id INT)  
BEGIN  
    SELECT * FROM billing WHERE student_id = stud_id;  
END $$  
DELIMITER ;
```

Experimental Results and Discussion:



Login to University Resource Management System

First Name:

Password:

Login

Student Portal

[Dashboard](#) [Enrolled Courses](#) [Results](#) [Registration](#) [Billing](#) [Profile](#) [Logout](#)

Welcome, Shakil Hossain

Advisor: Dr.Alamgir Hossain

Enrolled Courses

Course Name	Credits	Course Fee	Enrollment Date	Progress
Introduction to Programming	1	5000.00	2023-01-10	in process
Structured Programming	3	15000.00	2023-01-10	in process
Structured Programming LAB	1	5000.00	2023-01-10	in process
Discrete Mathematics	3	15000.00	2023-01-10	in process
Data Structures	3	15000.00	2023-01-10	complete
Data Structures LAB	1	5000.00	2023-01-10	in process
Digital Logic Design	3	15000.00	2023-01-10	in process

[Back to Dashboard](#)

Your Results

Course Name	Grade	Semester
Introduction to Programming	A	Spring 2023
Structured Programming	B+	Spring 2023
Structured Programming LAB	A-	Spring 2023
Discrete Mathematics	A	Spring 2023
Data Structures	B	Spring 2023
Data Structures LAB	A+	Spring 2023
Digital Logic Design	B+	Spring 2023
Introduction to Programming	A+	Summer 2024

[Back to Dashboard](#)

Available Courses

Course Name	Credits	Course Fee	Actions
Digital Logic Design LAB	1	5000.00	✓ Add ✗ Cancel
Object Oriented Programming	3	15000.00	✓ Add ✗ Cancel
Object Oriented Programming LAB	1	5000.00	✓ Add ✗ Cancel
Algorithms	3	15000.00	✓ Add ✗ Cancel
Algorithms LAB	1	5000.00	✓ Add ✗ Cancel
Computer Organization and Architecture	3	15000.00	✓ Add ✗ Cancel
Database Management System	3	15000.00	✓ Add ✗ Cancel
Database Management System LAB	1	5000.00	✓ Add ✗ Cancel
Automata and Theory of Computation	3	15000.00	✓ Add ✗ Cancel
Operating Systems	3	15000.00	✓ Add ✗ Cancel
Operating Systems LAB	1	5000.00	✓ Add ✗ Cancel

Billing

Course Name	Credits	Course Fee	Actions
Introduction to Programming	1	5000.00	Cancel
Structured Programming	3	15000.00	Cancel
Structured Programming LAB	1	5000.00	Cancel
Discrete Mathematics	3	15000.00	Cancel
Data Structures	3	15000.00	Cancel
Data Structures LAB	1	5000.00	Cancel
Digital Logic Design	3	15000.00	Cancel

Total Amount: 75000

[Back to Registration](#)

User Profile

Name: Shakil Hossain
Email: shakil.hossain@bdmail.com
Blood Group: O+
Permanent Address: Sylhet, Bangladesh
Mobile Number: 01710000003

Teacher Dashboard

[Assigned Courses](#) [Upload Grades](#) [Profile](#) [Logout](#)

Teacher Dashboard

Welcome to your dashboard. Use the navigation bar above to manage your courses and students.

Teacher Dashboard

[Assigned Courses](#) [Upload Grades](#) [Profile](#) [Logout](#)

Assigned Courses

Course ID	Course Name	Credits
1	Introduction to Programming	1
2	Structured Programming	3
3	Structured Programming LAB	1
4	Discrete Mathematics	3

Manage Students

Student ID	Name	Course Name	Actions
1	Shakil Hossain	Introduction to Programming	Upload Results
3	Tariq Rahman	Introduction to Programming	Upload Results
1	Shakil Hossain	Structured Programming	Upload Results
4	Sultana Nazneen	Structured Programming	Upload Results
6	Samiya Ahmed	Structured Programming	Upload Results
1	Shakil Hossain	Structured Programming LAB	Upload Results
2	Mina Begum	Structured Programming LAB	Upload Results
1	Shakil Hossain	Discrete Mathematics	Upload Results
5	Jahangir Alam	Discrete Mathematics	Upload Results

Upload Result

Grade

Semester

[Upload](#)[Cancel](#)

User Profile

Name: Dr. Alamgir Hossain**Email:** alamgir.hossain@bdmail.com**Blood Group:** O+**Permanent Address:** Dhaka, Bangladesh**Mobile Number:** 01710000013

Admin Dashboard

Welcome to the Admin Dashboard

You can manage users, courses, and other administrative tasks from here.

Manage Users

User ID	Full Name	Role	Actions	
1	Rahim Uddin	admin	Edit	Delete
2	Karim Chowdhury	admin	Edit	Delete
3	Shakil Hossain	student	Edit	Delete
4	Mina Begum	student	Edit	Delete
5	Tariq Rahman	student	Edit	Delete
6	Sultana Nazneen	student	Edit	Delete
7	Jahangir Alam	student	Edit	Delete
8	Samiya Ahmed	student	Edit	Delete
9	Rohit Biswas	student	Edit	Delete
10	Nazia Sultana	student	Edit	Delete
11	Rafiqul Islam	student	Edit	Delete

Edit User

First Name

Last Name

Email

Advisor

Add User

First Name

Last Name

Password

Role

Admin

Admin

Student

Faculty

Blood Group

Permanent Address

Mobile Number

Manage Courses

Course ID	Course Name	Credits	Course Fee	Faculty Name	Actions
1	Introduction to Programming	1	5000.00	Dr. Alamgir Hossain	Edit Delete
2	Structured Programming	3	15000.00	Dr. Alamgir Hossain	Edit Delete
3	Structured Programming LAB	1	5000.00	Dr. Alamgir Hossain	Edit Delete
4	Discrete Mathematics	3	15000.00	Dr. Alamgir Hossain	Edit Delete
5	Data Structures	3	15000.00	Dr. Nabila Ahmed	Edit Delete
6	Data Structures LAB	1	5000.00	Dr. Nabila Ahmed	Edit Delete
7	Digital Logic Design	3	15000.00	Dr. Nabila Ahmed	Edit Delete
8	Digital Logic Design LAB	1	5000.00	Dr. Nabila Ahmed	Edit Delete
9	Object Oriented Programming	3	15000.00	Dr. Nabila Ahmed	Edit Delete
10	Object Oriented Programming LAB	1	5000.00	Dr. Nabila Ahmed	Edit Delete
11	Algorithms	3	15000.00	Prof. Asad Rahman	Edit Delete

Discussion

During development, several decisions were made to ensure the system remains scalable and user-friendly. For example, real-time data sync was prioritized over full-page reloads to enhance the user experience. A modular folder structure allowed easier debugging and updates. The student registration process was simplified using dropdown selections linked directly to the database, reducing errors. Testing was done with various roles to ensure permissions were correctly enforced. The combination of PHP for logic and AJAX for real-time response ensured the system performed smoothly without excessive server load. Future updates may include reporting tools and notification systems.

Challenges and Solutions:

Challenge	Solution
SQL Injection Risk	Used parameterized queries with <code>mysqli_prepare()</code> and input validation
Session Hijacking	Implemented session ID regeneration and timeout logic
Complex User Access Roles	Implemented RBAC logic at both front and backend for consistent control
Live Updates Without Reloading Pages	Integrated AJAX-based data fetch and update mechanisms
Error Debugging in Large PHP	Modularized code and used try-catch blocks with error logs
Responsive UI for All Devices	Applied Bootstrap grid system and media queries for mobile-friendliness

Project Timeline:

Phase	Duration	Details
Requirement Gathering	1 week	Interviews, needs assessment, documentation
Design	1 week	Flowcharts, wireframes, and database schema
Implementation	2 weeks	Module development for each role-based section
Testing & Debugging	1 week	Unit testing, user testing, bug resolution
Documentation & Report	1 week	Final report, flowcharts, and system documentation

Conclusion:

URMS provides a much-needed digital transformation in managing university resources efficiently and securely. It improves communication between university departments and reduces administrative workload through automation. Role-based access ensures data is accessible only to authorized users. By addressing core issues like result management, billing, and course registration, the system brings transparency and accountability. Future versions could include push notifications, user messaging, mobile app integration, and reporting dashboards for analytics. The current system is ready for live deployment and can be scaled as per institutional needs.

References:

1. Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson.
Used for understanding relational database design and Entity-Relationship modeling.
2. Welling, L., & Thomson, L. (2009). *PHP and MySQL Web Development* (4th ed.). Addison-Wesley.
A foundational guide for developing dynamic, database-driven websites using PHP and MySQL.
3. Bootstrap Documentation.
<https://getbootstrap.com/docs/>
Referenced for responsive design elements, styling, and layout structure.